

ASSIGNMENT#2

FE EL

CLO3-(PBL)

COVERING PAGE

Subject: PH-112 **Max.Marks:** 05 **Type:** Individual

Title: The purpose of this PBL is to study quantum mechanics and photoelectric effect by determining plank's constant, work function and photon's speed in the given problem

Linkage to Course Learning Outcome			
CLO Number	Description of CLO	DOMAIN	Taxonomy level
CLO-3	Apply the concepts of Modern Physics to solve the problems.	Cognitive	3
Sr.no.	Problems Description		
1.	To determine the work function and Planck's Constant from the increasing relation between wavelength and Stopping Potential.		
2.	To determine the cut off wavelength for metal surface when illuminated by monochromatic light.		
Sr.no.	Constraints/Assumptions		
1.	Work function of Sodium and Silver is given that is 2.64ev and 4.73ev		
2.	Same monochromatic light strikes on both metal surfaces		
3.	Measure the K.E of electron from both the surfaces		
4.	From Work function we can calculate the cut off wavelength for both metal		
Sr.no.	Identification of Areas where use of computational/modern Tools usage is required.		
1.	Firstly require a monochromatic light source		
2.	For finding K.E and ϕ_o of electron, we use photoelectric apparatus and analytical software		
3.	Visualization of data collected as graph in Microsoft Excel Software		
4.	Report Writing in Microsoft word software		
Sr.no.	Expected Outcomes		
1.	Determination of (K.E) of electron from different metals and also its work function		
2.	Understanding by using calculations and its formula		
3.	Report writing with proper formatting		
Sr.no.	Complex Engineering Activities		
	Preamble: Complex activities means (engineering) activities or projects that some or all of the following characteristics listed below:		
	Attributes of CEA	Description	Apply
1.	Range of resources	Involve the use of diverse resources (for this purpose resources include money, equipment, materials, information and technologies).	Yes
2.	Level of Interaction	Requires solution of significant problems arising from interaction between wide ranging of conflicting technical, engineering and other issues	No
3.	Innovation	Involve creative use of engineering principles and research based knowledge in novel ways	No
4.	Consequences to society and environ	Have significant consequences in range of context, characterized by difficulty of prediction and mitigation	No
5.	Familiarity	Can extend beyond previous experience by applying principle base approach	Yes

Teacher Signature.....

Teacher Name.....

ASSIGNMENT#2 (CLO-3)
DEPARTMENT OF ELECTRONIC ENGINEERING

MAX.MARKS:05

QUESTION NO 1: The data in table was obtained by Robert A. Millikan (United States, 1868-1953), who was awarded the 1921 Nobel prize in **physics for his verification of the photoelectric effect**. Millikan's photoelectric data for lithium are:

Wavelength (nm)	433.9	404.7	365.0	312.5	253.5
Stopping potential (V)	0.55	0.73	1.09	1.67	2.57

- Make a plot of the table on graph paper with proper scaling
- Find the Planks constant
- Work function for lithium

QUESTION NO 2: Sodium and silver have work function of 2.64eV and 4.73eV, respectively.

- If the surfaces of both metals are illuminated with same monochromatic light which metal will give off photons with greater speed? How much faster will those electrons be?
- Calculate the cutoff wavelength of both metals?

INSTRUCTIONS:

- Submit the assignment in A4 sheet, with name and roll no. written on the very right side of the sheet
- Last date for the submission is **18-NOV -2024**

NED University of Engineering & Technology
Department of Electronic Engineering

Course Code & Title: PH-112
Assessment Rubric for PBL

Student's Name: Roll No:

Criterion	Level Of Attainment				
	Below Average (1)	Average (2)	Good (3)	Very Good (4)	Excellent (5)
Understanding the Problem	Didn't understand the problem	Preparation of data	Correct data collection points	Correct measurement orientation	Correct data collection
Analysis of data	No data table in MS Excel	Correct data table in MS Excel	Graphs in MS Excel	Discovering of relationship	Preparation of report
Timely Submission	No	----	----	----	Yes
Quality of Report	Covers no detail	Covers limited details	Covers required details	Covers full details	Presented in professional style

Total Score = _____ **Instructor's Signature:** _____